

Chapter 2: Literature Review

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LEO Low Earth Orbit

TLE Two Line Elements

JSPOC Joint Space Operations Center

I. INTRODUCTION

Space has been at the forefront of research and exploration for decades and much of modern technology has come to rely on the innovations and architecture that have been built to maintain it. The largest growing component of space innovation involves the deployment of Low Earth Orbit (LEO) satellites [1]. This greatly expanding front of new technology and ideas brings along its own challenges that range from security to new ways to track orbits [2, 3].

II. CUBESATS

A. Security

TABLE I
TWO-LINE ELEMENT SET FIELD DEFINITIONS

Line	Field	Content
Title	–	Satellite Name
1	1	Line Number
1	2	Satellite Catalog Number
1	3	Classification (U/C/S)
1	4	International Designator (Launch Year)
1	5	International Designator (Launch Number)
1	6	International Designator (Launch Piece)
1	7	Epoch Year
1	8	Epoch (Day and Fractional Day)
1	9	First Derivative of Mean Motion (rev/day ²)
1	10	Second Derivative of Mean Motion (rev/day ³)
1	11	B* Drag Term (Earth radii ⁻¹)
1	12	Ephemeris Type
1	13	Element Set Number
1	14	Checksum (modulo 10)
2	1	Line Number
2	2	Satellite Catalog Number
2	3	Inclination (deg)
2	4	Right Ascension of Ascending Node (deg)
2	5	Eccentricity
2	6	Argument of Perigee (deg)
2	7	Mean Anomaly (deg)
2	8	Mean Motion (rev/day)
2	9	Revolution Number at Epoch
2	10	Checksum (modulo 10)

III. NAVIGATION

LEO satellites use a variety of methods to assist in their own navigation and an organization's ability to track them from the ground. The most common methods used to track a satellite from the ground is through Two Line Elements (TLE)s. TLEs describe the orbital parameters, as seen in Table III, that are required for the tracking and locating of specific satellites during orbit [4]. They provide the specific time and location that each LEO satellite occupies to facilitate communication and to ensure that no satellite impacts another. The recordings of a satellite's orbit are taken through observation at Joint Space Operations Center (JSPOC) and provide consistent snapshot updates of currently recorded TLEs every 8 hours [4]. A majority of these satellites have a mean motion of 90 minutes with an altitude (Circular) of 281.49 km [5]. This provides only a few minutes in which the satellite can be contacted before it is out of range and continues until its next orbit. These minimal contact intervals make it vitally important to accurately predict when contact can be made and what communications are of import to send during this window.

A. Alternative Navigation

B. Magnetometer And Gravitometer

Alternative methods of navigation and orbit determination continue to be explored to provide more robust and accurate systems. A newly explored source of navigation is the use of satellite sensors, like magnetometer and gravimeter, in conjunction with accurate modeling to derive orbit parameters without specific ground station interference [6, 7].

C. Other Methods

D. ML Models

E. Machine Learning

F. Neural Networks

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